

Product Requirement Document (PRD) & Technical Specification

Project Name: Shailaja IAS E-Learning Platform

Version: 1.0 (Integrated Final)

Date: January 2026

1. Executive Summary

The Shailaja IAS platform is a high-performance E-Learning ecosystem for UPSC aspirants. The system prioritizes high-volume content delivery (Current Affairs) and a rigorous assessment engine (Test Series). To ensure long-term viability, the architecture utilizes a Hybrid Video Engine for cost optimization and a Custom-Built Admin Portal designed for rapid data entry (Bulk CSV).

2. User Personas

- 1. Aspirant (User):** Self-identifies as Beginner, Working Professional, or Veteran to receive tailored course recommendations.
- 2. Admin:** Manages thousands of questions, daily news articles, and video course hierarchies.
- 3. Evaluator:** A specialized role focused on the evaluation and re-upload of subjective Mains answer scripts.

3. Functional Modules

Content unchanged. Refer to previous sections.

4. High-Level Design (HLD)

Technology Stack

Frontend: Next.js (SSR/SSG).

Backend: Node.js + Express.

Database: PostgreSQL or MongoDB.

Caching: Redis.

Storage: AWS S3 + CloudFront.

5. Implementation Roadmap

Assumption: ~18 hours/week. Phase details remain unchanged.

7. Infrastructure Cost Summary

Item	Service Provider	Monthly Estimate
Frontend	Vercel (Pro)	\$20
Backend API	AWS App Runner / EC2	\$15 – \$25
Database	MongoDB Atlas	\$15 – \$30
Video (Secure)	Bunny.net Stream	\$5 – \$15
Storage & CDN	AWS S3 + CloudFront	\$5 – \$10
Misc (Redis/DNS)	Upstash / Route53	\$10

TOTAL		\$70 – \$110 / month
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8. Scalability Roadmap

Users	Concurrent Users	Monthly Budget	Infrastructure Focus
1,000 (Launch)	100	\$70 – \$110	Shared MongoDB Atlas, Vercel Pro, Small I
5,000	500	\$180 – \$300	Dedicated MongoDB (M10), Redis caching
25,000	2,500	\$650 – \$1,200	MongoDB sharding, Multi-region CDN, Auto
50,000	5,000	\$1,500 – \$3,500+	Enterprise MongoDB clusters, Advanced W